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MINI PROJECT REPORT

ONLINE VOTING SYSTEM

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Introduction:

Voting is one of the most important rights in a democratic system, allowing people to choose their representatives and participate in decision-making. Traditional voting methods, such as paper ballots and Electronic Voting Machines (EVMs), have been widely used for many years. However, these methods often require significant time, manpower, and resources. They may also involve long queues, manual verification, and delays in counting votes and announcing results.

The Online Voting System is a web-based application developed to simplify and modernize the election process. It enables eligible voters to cast their votes securely through the internet using a computer or mobile device. The system provides a convenient and efficient way to conduct elections while reducing paperwork and minimizing human errors.

The application consists of two main modules: the Administrator Module and the Voter Module. The administrator is responsible for managing voters, candidates, and elections, while voters can log in using their credentials, view the list of candidates, and cast their vote. The system ensures that each registered voter can vote only once, helping to prevent duplicate voting and maintain the integrity of the election.

One of the major advantages of the Online Voting System is its ability to generate election results automatically. Since votes are stored digitally in a database, the system can count them quickly and accurately, reducing the time required to declare results. This improves efficiency, transparency, and reliability compared to traditional voting methods.

The project is developed using modern web technologies such as HTML, CSS, JavaScript, PHP, and MySQL. HTML, CSS, and JavaScript are used to create an interactive and user-friendly interface, while PHP handles the application logic and MySQL securely stores voter, candidate, and election data.

The Online Voting System is suitable for colleges, universities, schools, clubs, and organizations that conduct elections regularly. It reduces administrative work, lowers operational costs, saves time, and provides a secure and convenient voting experience for users. With future enhancements such as OTP verification, biometric authentication, and mobile application support, the system can be made even more secure and accessible.

Overall, the Online Voting System offers a reliable, transparent, and efficient solution for conducting elections digitally. It improves the voting process by ensuring fairness, reducing manual effort, and delivering fast and accurate election results.

Project Objective:

The primary objective of the Online Voting System is to develop a secure, efficient, and user-friendly web application for conducting elections digitally. The system aims to simplify the voting process, improve transparency, and reduce the time and effort required in traditional election methods.

The specific objectives of the project are:

- To develop a secure online platform for conducting elections.
- To allow eligible voters to register and log in using authenticated credentials.
- To ensure that each voter can cast only one vote during an election.
- To provide an easy-to-use interface for both voters and administrators.
- To enable administrators to manage voters, candidates, and election details efficiently.
- To automate the vote counting process and generate accurate election results instantly.
- To reduce paperwork, manual errors, and the overall cost of conducting elections.
- To maintain the confidentiality, integrity, and security of voting data.
- To improve transparency and reliability in the election process.
- To provide a faster and more convenient voting experience for users.

Major Objectives of the Project:

The major objective of the **Online Voting System** is to develop a secure, reliable, and efficient web-based application that simplifies the election process by allowing eligible voters to cast their votes electronically. The system is designed to reduce manual effort, improve transparency, and provide accurate election results in a shorter time.

The major objectives of the project are:

- To provide a secure and user-friendly online voting platform.
- To authenticate voters and ensure that only eligible users can participate in the election.
- To prevent duplicate voting by allowing each voter to cast only one vote.
- To enable administrators to manage voters, candidates, and election activities efficiently.

- To automate the vote counting process and generate accurate results instantly.
- To reduce the time, cost, and paperwork involved in traditional voting methods.
- To maintain the confidentiality, integrity, and security of voting data.
- To improve transparency, reliability, and fairness throughout the election process.
- To provide a scalable system that can be used for elections in colleges, organizations, and other institutions.

System Modules:

The Online Voting System is designed using different functional modules, where each module performs a specific task. These modules work together to ensure that the voting process is secure, efficient, and easy to manage. The modular structure also makes the system easier to maintain, update, and expand in the future.

1. Administrator Module

The Administrator Module is the central part of the system and is responsible for managing all election-related activities. The administrator has complete control over the application, including the management of voters, candidates, and election details. This module also allows the administrator to monitor the voting process and publish the final election results.

Functions:

- Secure administrator login
- Add, update, and delete voter details
- Add, update, and delete candidate details
- Create and manage elections
- Monitor voting status
- View and publish election results

2. Voter Module

The Voter Module enables registered users to participate in the election through a secure login. After authentication, voters can view the list of candidates, cast their vote, and receive confirmation. The system ensures that each voter is allowed to vote only once.

Functions:

- Voter login

- View candidate information
- Cast vote securely
- Receive vote confirmation
- Logout from the system

3. Authentication Module

The Authentication Module is responsible for verifying the identity of users before granting access to the system. It protects the application by ensuring that only authorized administrators and registered voters can log in. This module also manages user sessions to maintain system security.

Functions:

- User authentication
- Login verification
- Password validation
- Session management
- Prevent unauthorized access

4. Voting Module

The Voting Module manages the entire voting process. It displays the list of candidates, records the selected vote, and stores it securely in the database. The module prevents duplicate voting by ensuring that each registered voter can submit only one vote during an election.

Functions:

- Display candidates
- Record votes securely
- Prevent multiple voting
- Store voting information
- Maintain vote confidentiality

5. Result Management Module

The Result Management Module automatically counts the votes stored in the database and generates the final election results. It provides accurate and quick results, reducing the need for manual counting and minimizing the chances of errors.

Functions:

- Automatic vote counting

- Generate election results
- Display winning candidate
- View election statistics
- Publish final results

System Architecture:

The Online Voting System follows a web-based client–server architecture that enables secure communication between users and the database through the application server. The architecture is divided into four main layers: the User Layer, Presentation Layer, Application Layer, and Data Layer. Each layer performs a specific function to ensure the smooth and secure operation of the system.

The **User Layer** consists of two types of users: the **Administrator** and the **Voter**. The administrator manages the election process by creating elections, adding candidates, registering voters, monitoring voting activities, and publishing results. Registered voters access the system through a web browser, log in using their credentials, and cast their votes securely.

The **Presentation Layer** provides the graphical user interface (GUI) of the application. It includes login pages, registration forms, the administrator dashboard, candidate lists, voting pages, and result pages. This layer is designed to be simple, responsive, and user-friendly, allowing users to interact with the system easily.

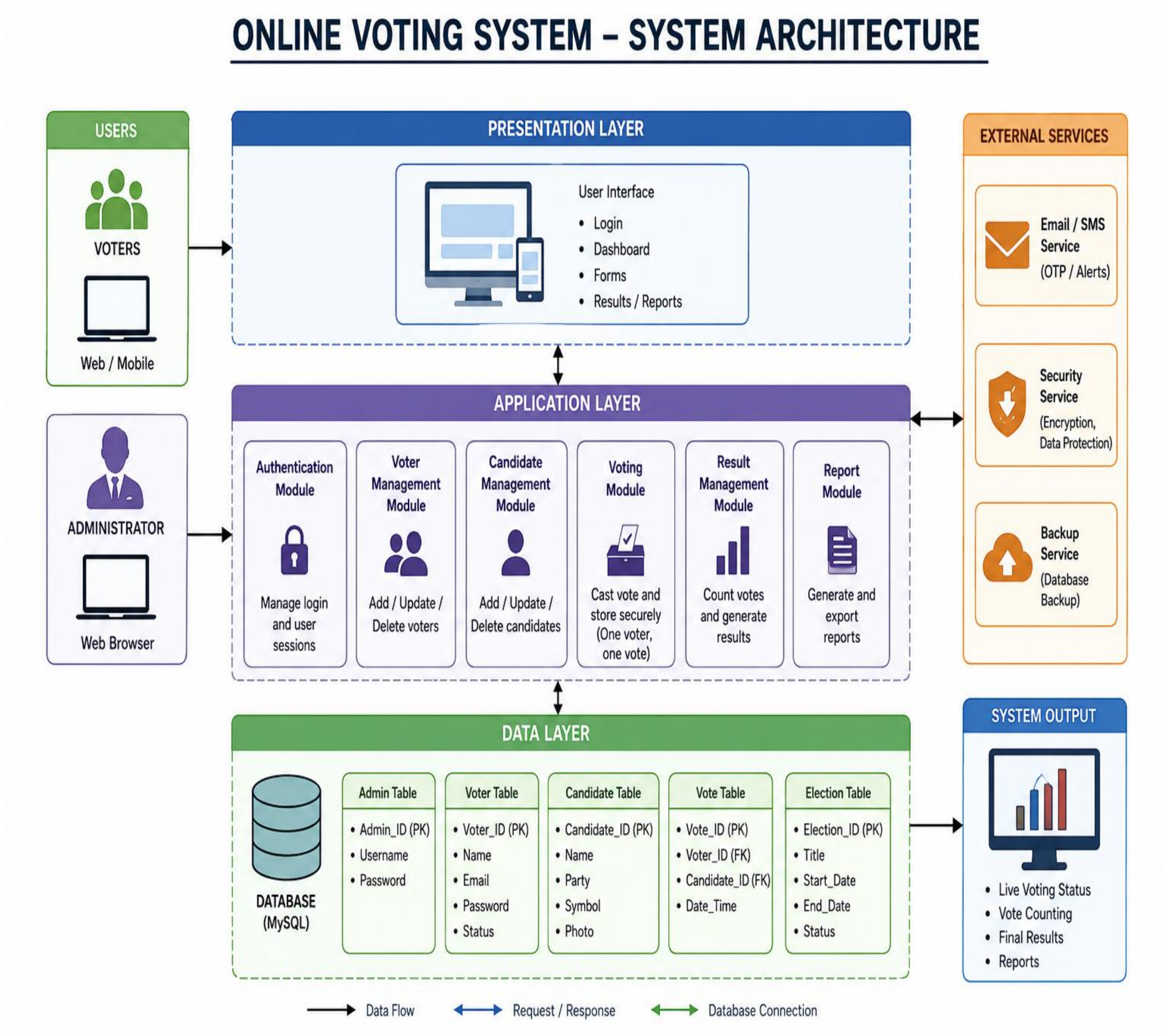
The **Application Layer** contains the core business logic of the system. It processes user requests, authenticates users, validates voter eligibility, manages candidates and elections, records votes, prevents duplicate voting, counts votes automatically, and generates election results. This layer acts as the bridge between the user interface and the database.

The **Data Layer** is responsible for storing and managing all election-related information in a MySQL database. It maintains separate tables for administrators, voters, candidates, elections, and votes. The database ensures data consistency, integrity, and secure storage of information throughout the election process.

When a voter submits a vote, the request is sent to the application layer, where the system verifies the voter's identity and checks whether the voter has already voted. If the verification is successful, the vote is securely stored in the database. After the election ends, the system automatically counts all valid votes and displays the final results to the administrator.

Overall, the proposed system architecture provides a secure, reliable, and scalable platform for conducting online elections. It minimizes manual intervention, improves transparency, ensures data security, and enables accurate and efficient vote management.

System Architecture-Online Voting System



Implementation:

1. Administrator Login Module (PHP)

PHP

```php
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Online Voting System</title>

    <link rel="stylesheet" href="css/style.css">
</head>
<body>

<div class="container">

    <div class="card">
        <h1>ONLINE VOTING SYSTEM</h1>
        <p>Welcome to the Online Voting System</p>

        <div class="buttons">
            <a href="admin/login.php">
                <button class="btn admin">Admin Login</button>
            </a>

            <a href="voter/login.php">
                <button class="btn voter">Voter Login</button>
            </a>
        </div>
    </div>

</div>

</body>
</html>
```

2. Voter Module:

The Voter Module enables registered users to log in using their credentials and participate in the election. After successful authentication, the voter can view the list of candidates and cast a vote. The system verifies that the voter has not voted previously before recording the vote in the database. A confirmation message is displayed after the vote is submitted successfully.

PHP

```php
<?php
// Database Connection

$servername = "localhost";
$username   = "root";
$password   = "";
$database   = "online_voting";

// Create Connection
$conn = mysqli_connect($servername, $username, $password, $database);

// Check Connection
if (!$conn) {
    die("Database Connection Failed: " . mysqli_connect_error());
}

// Set Character Encoding
mysqli_set_charset($conn, "utf8");
?>
```

3. Authentication Module

The Authentication Module is responsible for validating user credentials during login. It checks the entered username and password against the database and creates a secure user session upon successful authentication. Invalid login attempts are rejected, preventing unauthorized users from accessing the system.

Authentication Module (Sample Code)

The Authentication Module verifies the user's login credentials and grants access only to authorized users.

```
<?php
session_start();
include("../db.php");

$username = $_POST['username'];
$password = $_POST['password'];

$sql = "SELECT * FROM admin
        WHERE username='$username'
        AND password='$password'";

$result = mysqli_query($conn, $sql);

if(mysqli_num_rows($result) > 0){
    $_SESSION['admin'] = $username;
    header("Location: dashboard.php");
}
else{
    echo "Invalid Login!";
}
?>
```

4. Result Management Module

The Result Management Module automatically counts the votes stored in the database after the election ends. It calculates the total votes received by each candidate, identifies the winner, and displays the election results in an organized format. This module eliminates manual vote counting and ensures accurate and timely result generation.

```php
<?php
include("../db.php");

$sql = "SELECT candidate_name, COUNT(*) AS total_votes
        FROM votes
        GROUP BY candidate_name";

$result = mysqli_query($conn, $sql);

while($row = mysqli_fetch_assoc($result)){
    echo $row['candidate_name'] . " : " . $row['total_votes'] . " Votes <br>";
}
?>
```

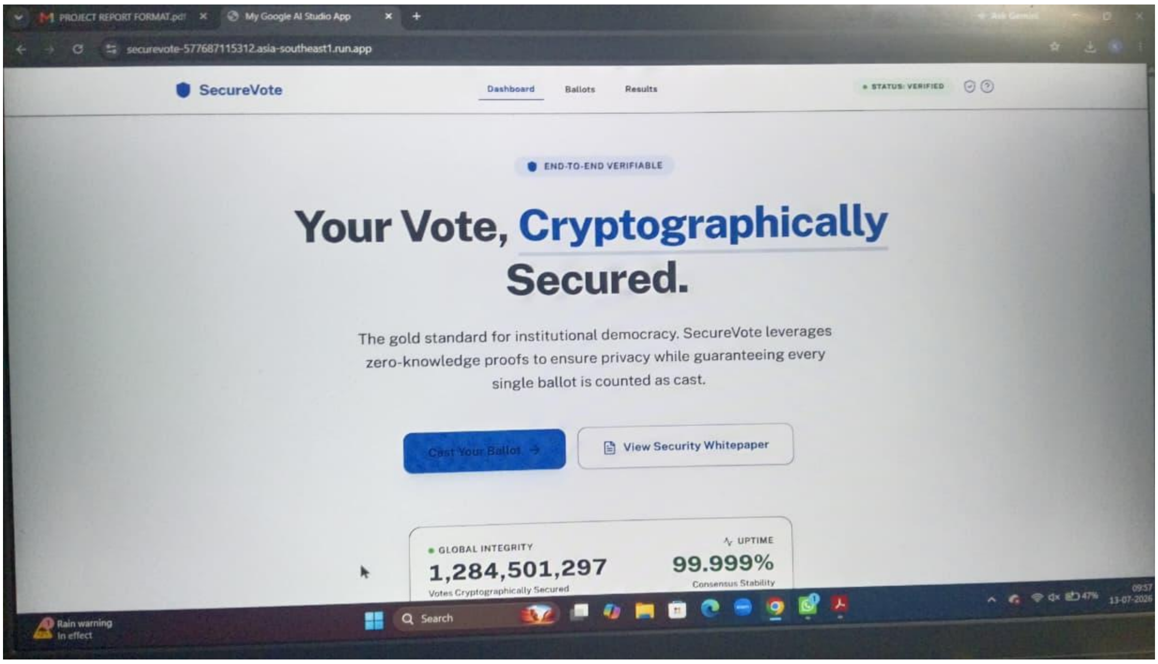
6. Database Module

The Database Module is implemented using MySQL and stores all information related to administrators, voters, candidates, elections, and votes. PHP scripts are used to establish a secure connection with the database and perform operations such as data insertion, retrieval, updating, and deletion using SQL queries.

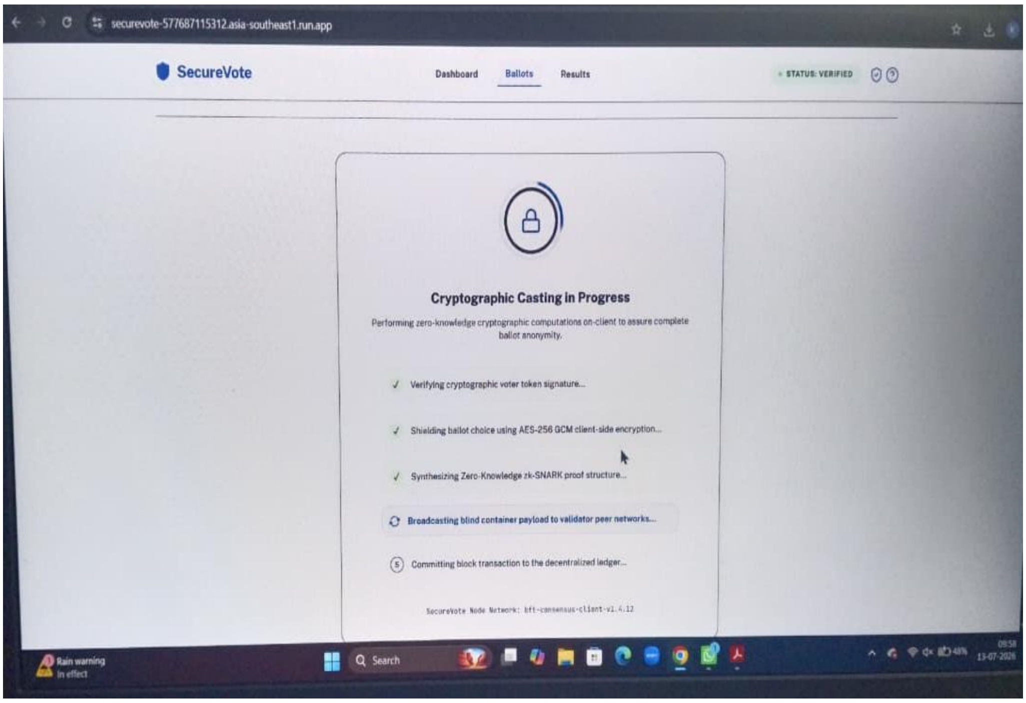
Overall, the modular implementation of the Online Voting System improves the efficiency, security, and maintainability of the application. Each module performs a dedicated function while working together to provide a reliable and user-friendly online voting platform.

```php
<?php
$host = "localhost";
$user = "root";
$pass = "";
$db   = "online_voting";

$conn = mysqli_connect($host, $user, $pass, $db);

if(!$conn){
    die("Database Connection Failed");
}
?>
```

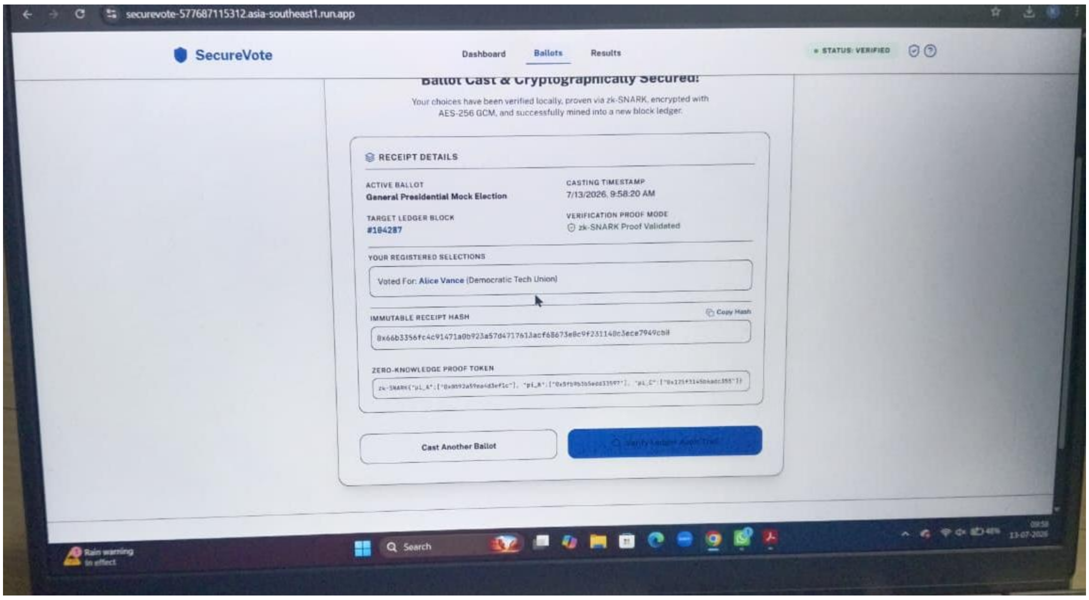
Home Page: Screenshots



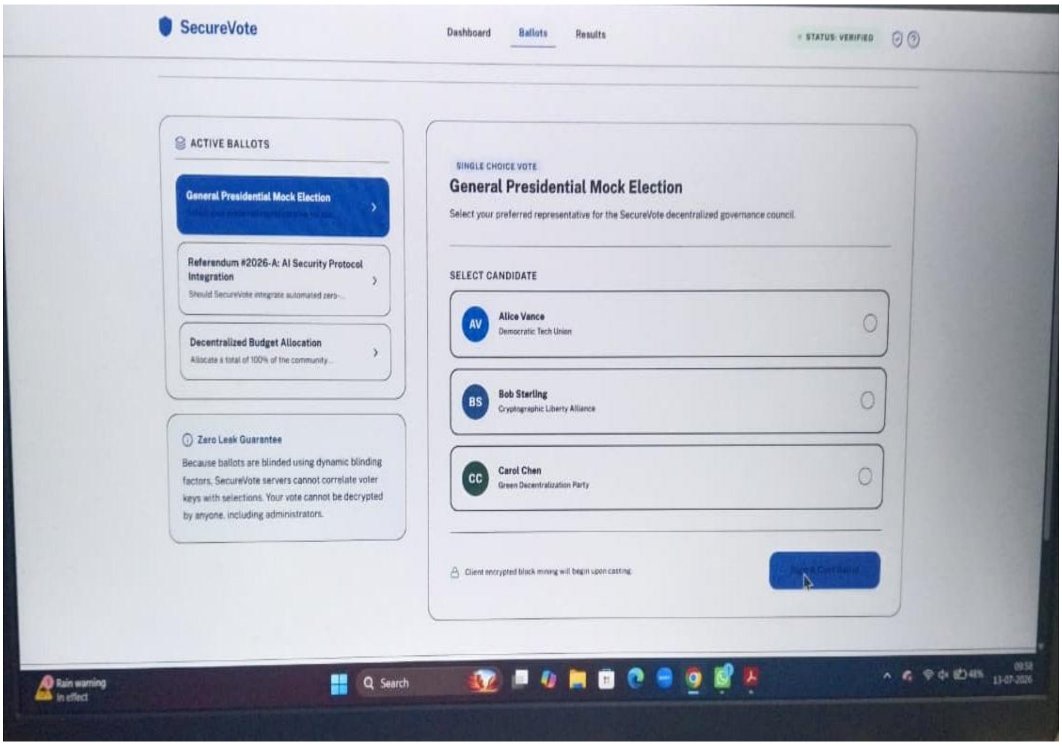
Ballot Page(Section-1):



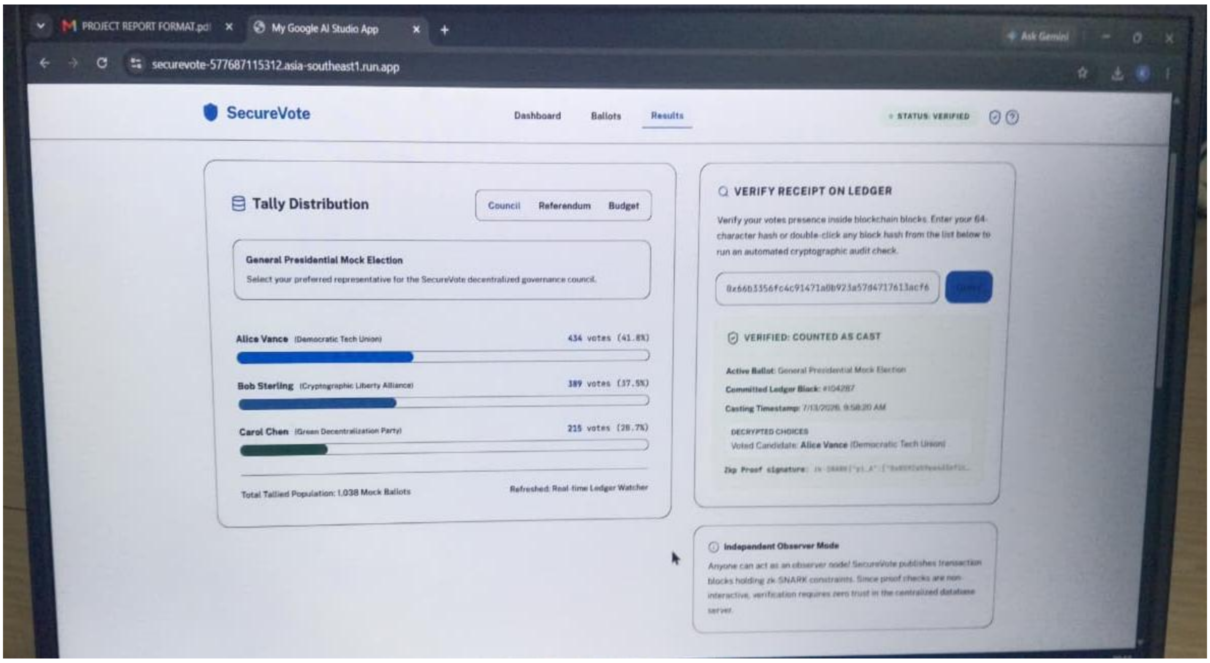
Ballot Page (Section-2):



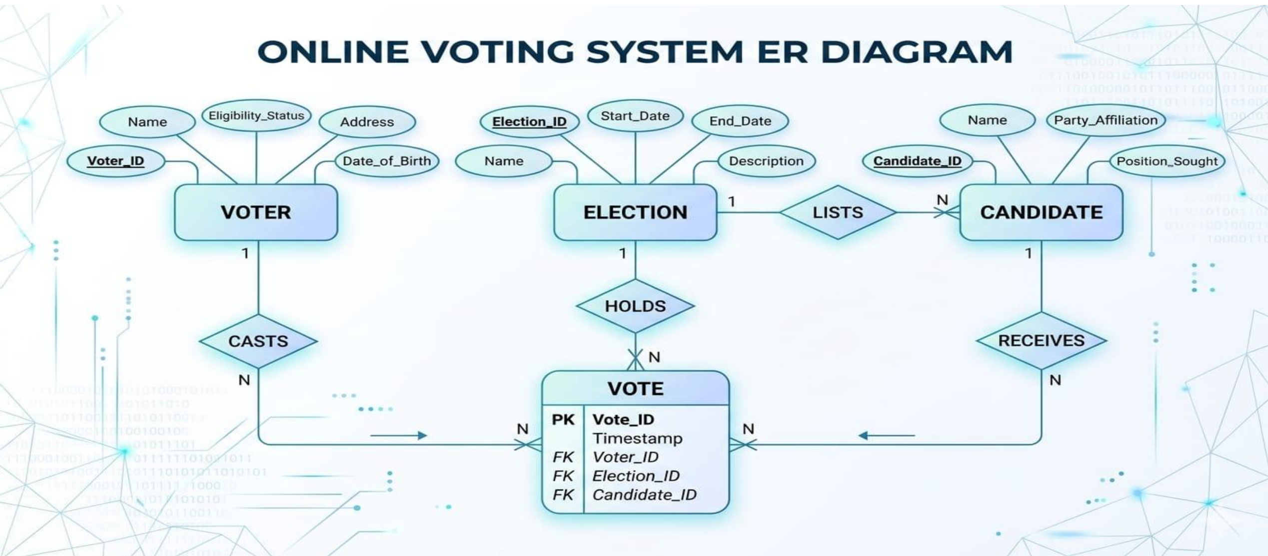
Ballot Page (Section-3):



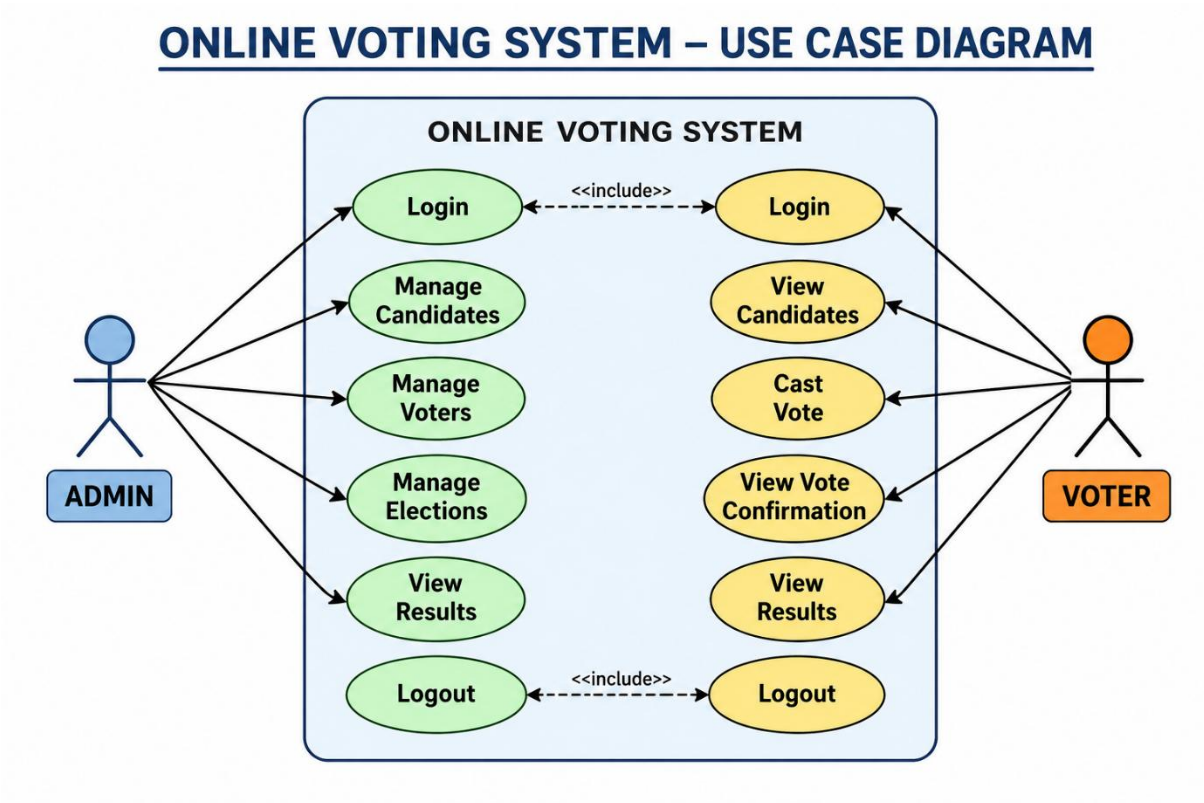
Result Page:



ER Diagram:



Case Diagram:



Software Used:

- **HTML5** – Used to design the structure of web pages.
- **CSS3** – Used to create an attractive and responsive user interface.
- **JavaScript** – Used for client-side validation and interactive features.
- **PHP** – Used as the server-side scripting language to implement the application's business logic.
- **MySQL** – Used to store voter, candidate, election, and voting information securely.
- **XAMPP** – Provides the Apache web server, PHP interpreter, and MySQL database required to run the application locally.
- **Visual Studio Code** – Used for writing, editing, and debugging the source code.
- **phpMyAdmin** – Used to create and manage the MySQL database.
- **Google Chrome / Microsoft Edge** – Used for testing and accessing the web application during development.

Technology Stack:

- **Frontend:** HTML5, CSS3, JavaScript
- **Backend:** PHP
- **Database:** MySQL
- **Session Handling:** PHP Sessions
- **Web Server:** Apache (XAMPP)
- **IDE:** Visual Studio Code
- **Database Management Tool:** phpMyAdmin
- **Web Browser:** Google Chrome / Microsoft Edge
- **Version Control:** Git & GitHub (Optional)
- **Architecture:** Client–Server Architecture (Three-Tier Architecture)

Advantage & Disadvantage:

ADVANTAGES AND DISADVANTAGES OF ONLINE VOTING SYSTEM

ADVANTAGES	DISADVANTAGES
✓ The voting process is faster and more efficient.	✗ Requires a stable internet connection and digital infrastructure.
✓ Voters can cast their votes from any location with an internet connection.	✗ Risk of hacking, cyber attacks, and data breaches.
✓ Results can be calculated and published instantly.	✗ Technical issues or system failures may disrupt the voting process.
✓ Reduces the use of paper, promoting a green and eco-friendly environment.	✗ Some voters may not be familiar with technology.
✓ Lower operational and administrative costs compared to traditional voting.	✗ Dependence on electronic devices, which may not be accessible to everyone.
✓ Improves accuracy and reduces human errors in counting votes.	✗ Potential for software bugs or glitches.
✓ Secure authentication ensures only eligible voters can participate.	✗ Ensuring complete transparency and trust can be challenging.
✓ Provides a digital record, making data storage and retrieval easier.	✗ Initial setup and maintenance costs can be high.

Conclusion:

The **Online Voting System** project successfully demonstrates a secure, efficient, and user-friendly approach to conducting elections through a web-based platform. The system simplifies the voting process by enabling eligible voters to cast their votes online while ensuring that each voter can vote only once. It also provides administrators with the tools to manage voters, candidates, elections, and results effectively.

The application reduces manual effort, minimizes paperwork, and provides accurate vote counting with faster result generation. By using technologies such as **HTML, CSS, JavaScript, PHP, and MySQL**, the system offers a reliable solution for conducting elections in educational institutions, organizations, and other small to medium-scale environments.

Overall, the Online Voting System improves transparency, security, and efficiency in the election process. It serves as a modern alternative to traditional voting methods and can be further enhanced with advanced security features such as biometric authentication, OTP verification, and encryption to support larger and more secure election environments.

References:

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[4] P. Deitel and H. Deitel, *Internet and World Wide Web: How to Program*, 5th ed. Pearson Education, 2012.

[5] Oracle Corporation, *MySQL 8.0 Reference Manual*. Available: https://dev.mysql.com/doc/

GITHUB LINK:

https://github.com/kavipriya9966-AIDS/Online-Voting-System-VSB

THANK YOU